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Regioselectivity

Connections

➡ Building on

- Chemoselectivity [ch23](#)
- Electrophilic aromatic substitution [ch21](#)
- Addition to alkenes [ch19](#)
- Substitution at saturated C [ch15](#)
- Electrophilic alkenes and nucleophilic aromatic substitution [ch22](#)

Arriving at

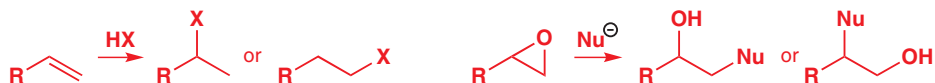
- Selectivity of a new kind decided by mechanism
- Reagent and substrate are both important
- Controlling the arrangement of aromatic substituents
- How to get *ortho* selectivity: ortholithiation and sulfonation
- Radical as well as ionic reactions
- Reactions of allylic compounds
- Revisiting conjugate addition

➡ Looking forward to

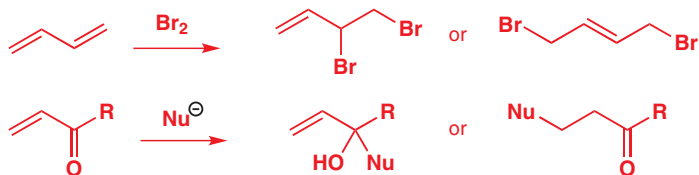
- Stereoselectivity [ch32–ch33](#)
- Enol and enolate reactions [ch25 & ch26](#)
- Reactions and synthesis of heterocycles [ch29 & ch30](#)
- Radical reactions [ch37](#)

Introduction

We met *chemoselectivity*—*which* group reacts—in the last chapter. Chemoselectivity means that there are two separate functional groups and that a reagent must choose between them. By contrast, *regioselectivity* implies that there is one functional group that can react in two different places and a reagent must choose where to react. Simple examples include addition of HX to an alkene (Chapter 19) and nucleophilic attack on the epoxide derived from that alkene (Chapter 15).

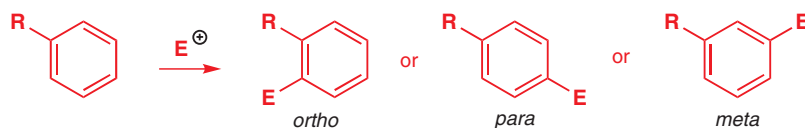


It might also mean that two functional groups are combined in a single conjugated system that can again react in two (or more) places. Examples include the addition of bromine to dienes (two conjugated alkenes) and addition of a nucleophile to a conjugated carbonyl compound (carbonyl group conjugated to an alkene).



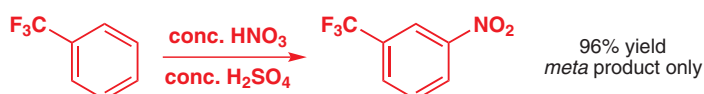
Online support. The icon  in the margin indicates that accompanying interactive resources are provided online to help your understanding: just type www.chemtube3d.com/clayden/123 into your browser, replacing **123** with the number of the page where you see the icon. For pages linking to more than one resource, type **123-1**, **123-2** etc. (replacing **123** with the page number) for access to successive links.

The choice between *ortho/para* and *meta* substitution when an electrophile attacks a benzene ring (Chapter 21) is also a matter of regioselectivity. We shall discuss all these examples in further detail in this chapter, and extend these ideas to new reactions as well.

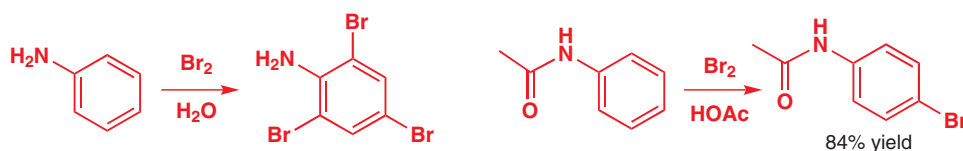


Regioselectivity in electrophilic aromatic substitution

We start with electrophilic aromatic substitution. It was established in Chapter 21 that an electron-donating substituent favours *ortho/para* and an electron-withdrawing substituent favours *meta* substitution. Although *meta* substitution is usually slower than *ortho/para* substitution (because electron-withdrawing groups deactivate the ring), it usually gives the *meta* product alone.



Most reactions of benzene rings with electron-donating substituents give *ortho/para* mixtures and, if the substituent is very electron-donating, may lead to both *ortho* and *para* substitution in the same molecule. Control in favour of the *para* product can usually be achieved by reducing the reactivity of the substituent and increasing its size.



Of course, if the *para* position is blocked, *ortho* substitution is the only option, and we will come back to the idea of blocking substituents shortly. But there is a general way of directing electrophiles to the *ortho* position using activation by metallation.

➡ All these examples are drawn from Chapter 21.



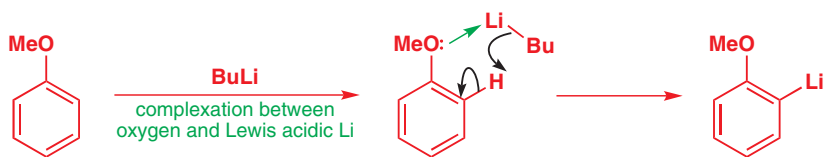
Making organometallics by deprotonating aromatic rings: ortholithiation

Look at the reaction below: butyllithium deprotonates an sp^2 hybridized carbon atom to give an aryllithium. It works because the protons attached to sp^2 carbons are more acidic than protons attached to sp^3 carbons (although they are a lot less acidic than alkyne protons).

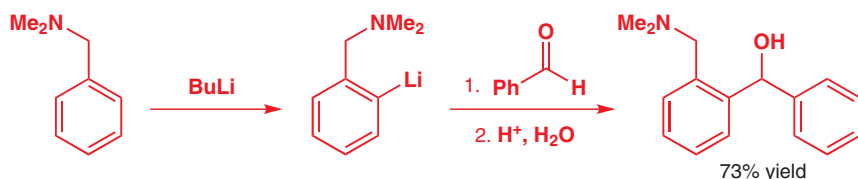


But there must be another factor involved to account for exclusive *ortho* substitution, which is after all the most hindered site. The functional group containing oxygen (sometimes nitrogen) is next to the proton to be removed. This functional group 'guides' the butyllithium, so that it attacks the adjacent protons. It does this by forming a complex with the Lewis acidic lithium atom, much as ether solvents dissolve Grignard reagents by complexing

their Lewis-acidic metal ions. This mechanism means that it is only the protons *ortho* to the functional group that can be removed, and the reaction is known as an **ortholithiation**.



The example below shows ortholithiation, activated by the nitrogen atom of a tertiary amine, being used to make a new C–C bond. Here it is the nitrogen atom that directs attack of the butyllithium, again by complexation with the Li atom.

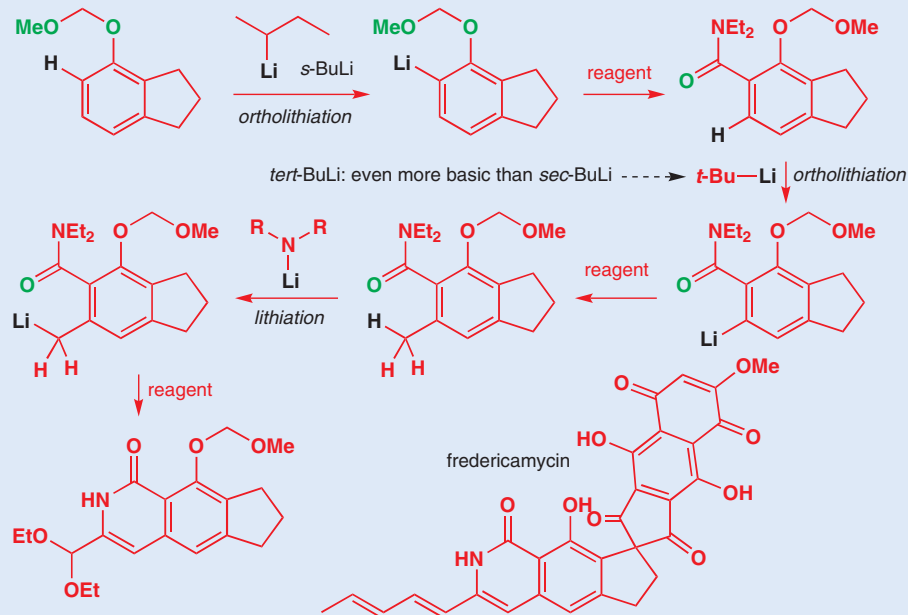


■ Compare the methods for making organolithiums or Grignard reagents that you met in Chapter 9—most of them rely on formation of an organometallic by reduction of an alkyl or aryl halide.

Ortholithiation is a useful way of making reactive organometallics because the starting material does not need to contain a halogen atom. But it is much less general than the other ways we have told you about for making organolithiums, as there are rather tight restrictions on what sorts of groups the aromatic ring must carry. The best ortholithiation-directing substituents have lone pairs to donate electrons to Li and are also electronegative so they withdraw electrons from the benzene ring and help stabilize the anion forming at the *ortho* position.

Fredericamycin

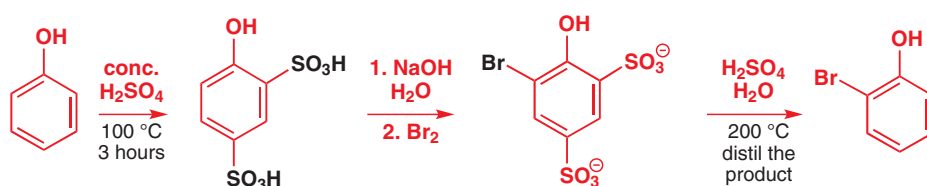
Fredericamycin is a curious aromatic compound extracted in 1981 from the soil bacterium *Streptomyces griseus*. It is a powerful antibiotic and antitumour agent, and its structure is shown below. The first time it was made in the laboratory, in 1988, the chemists in Boston started their synthesis with three consecutive lithiation reactions: two are ortholithiations, and the third is slightly different. You needn't be concerned about the reagents that react with the organolithiums; just look at the lithiation reactions themselves. In each one, one or more oxygen atoms (colour-coded green) directs a strongly basic reagent to remove a nearby proton (colour-coded black). As it happens, none of the steps uses *n*-BuLi itself, but instead its more reactive cousins, *sec*-BuLi and *tert*-BuLi (see the table on p. 186). The third lithiation step uses a different kind of base related to LDA, made by deprotonating an amine ($\text{p}K_{\text{a}}$ about 35). The black proton removed in this third lithiation is more acidic because it is next to an aromatic ring.



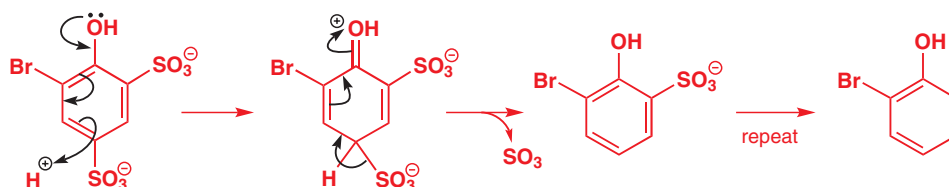
Sulfonation may lead to *ortho* selectivity without lithiation

We introduced sulfonation in Chapter 21 but have left detailed discussion until now because sulfonation has some features that make it more interesting than first meets the eye. One important difference between sulfonation and other examples of electrophilic substitution is that sulfonation is *reversible*. Heating an arenesulfonic acid causes it to decompose with loss of gaseous SO_3 .

Here's an example of how we can exploit this to gain control of regioselectivity without resorting to lithiation. In stage 1 the phenol is sulfonated twice—the first sulfonic acid group (which adds *para* to the OH group) is electron-withdrawing and deactivates the ring, making the introduction of the second group (which goes *ortho* to the OH and *meta* to the first sulfonic acid) harder and that of the third group harder still, which is why we can isolate the disulfonated phenol.

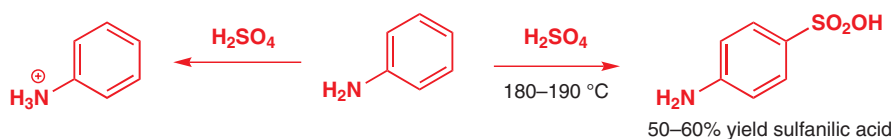


In the second stage, the bromination, the OH directs to the *ortho* and *para* positions, but only one *ortho* position is vacant, so the bromine attacks there. Sodium hydroxide is needed to deprotonate the sulfonic acid groups to make them less deactivating. The sulfonation reaction is reversible, and in the third stage it is possible to drive the reaction over to the products by distilling out the relatively volatile 2-bromophenol at high temperature. The loss of SO_3 involves attack of H^+ on the aromatic ring.



Overall, we have succeeded in making 2-bromophenol where direct bromination of phenol itself would have given (at low temperatures) mainly *p*-bromophenol and at higher temperatures, 2,4,6-tribromophenol. The sulfonic acid groups are useful reversible blocking groups.

The same method can be used with anilines because *para*-sulfonation of aromatic amines is possible. This seems surprising because in sulfuric acid essentially all the amine will be protonated. You might expect the resulting ammonium ion to react in the *meta* position (because NH_3^+ is no longer electron-rich) but instead the *para*-sulfonic acid (sulfanilic acid) is formed. At the high temperature of the reaction, it is probable that any *meta*-substituted product reverts to the starting material, while the *para*-sulfonic acid accumulates because it is stabilized by delocalization and is less hindered.



Regioselective reactions of naphthalene

We introduced you to the 10-electron aromatic system of naphthalene in Chapter 7. As you would expect, it undergoes electrophilic aromatic substitution with the same reagents you met in Chapter 21, but the regioselectivity of its reactions is of a different type to the *ortho*, *meta*,

The sulfonating agent

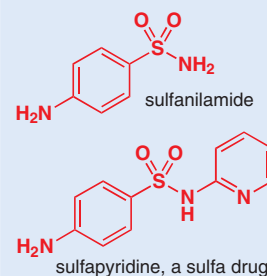
The exact nature of the electrophile in sulfonation reactions seems to vary with the amount of water present. Certainly for oleum (fuming sulfuric acid, that is concentrated sulfuric acid with added sulfur trioxide) and solutions of sulfur trioxide in organic solvents, the electrophile is sulfur trioxide itself, SO_3 . With more water around, H_3SO_4^+ and even $\text{H}_2\text{S}_2\text{O}_7$ have been suggested.

■ You might want to consider *why* sulfonation is reversible at high temperature in the light of our discussion of entropy and temperature on p. 248.

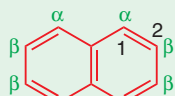
■ The reversibility of sulfonation with sulfuric acid may also account for the higher yield of *para* product in the sulfonation of toluene with H_2SO_4 as compared with ClSO_3H (p. 485).

Sulfa drugs

The product is important because the amides derived from it (sulfanilamides) were the first antibiotics, the sulfa drugs.



■ In Chapter 7 we pointed out that the middle bond is shorter than the rest, and for this reason we suggest you draw naphthalene with a double bond in this position—it makes mechanistic explanations more realistic.



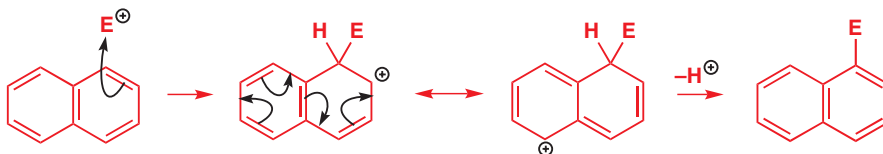
naphthalene has two sorts of ring positions

➡ Unactivated rings such as benzene need a Lewis acid to react with bromine: see p. 474.

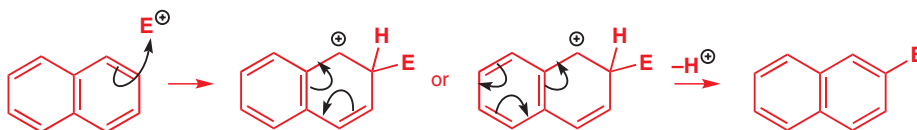
■ Another way of looking at the difference between these two delocalized cations is that the first can be shown delocalized into the double bond without disrupting the remaining aromatic ring; in the second, all other representations of delocalized structures must lose the aromatic ring.

para selectivity we have been talking about. Naphthalene has 10 carbons: two form the ring junction, and aren't available for substitution reactions, and the other eight are of just two types α (the 1-position, next to the ring junction) and β (the 2-position).

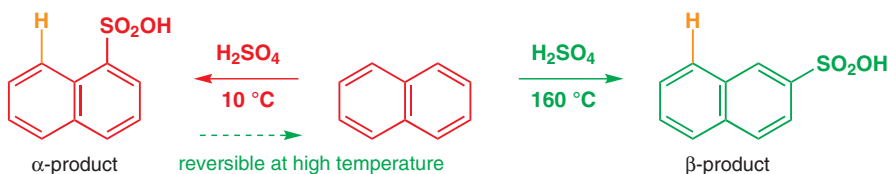
Electrophilic substitution on naphthalene normally occurs at a site next to the ring junction (α). This is because the HOMO has its largest coefficient at this atom, but you can rationalize the result by looking at the long, linear delocalization in the resulting cation, which can be represented by a single train of arrows. This extended conjugation makes naphthalene more nucleophilic than benzene. So, bromination occurs at the α -position in good yield even without a Lewis acid.



Reaction at the other position (β) is less favourable as the intermediate cation is cross-conjugated. The cation delocalizes into both rings, but no long linear chain of arrows is possible.

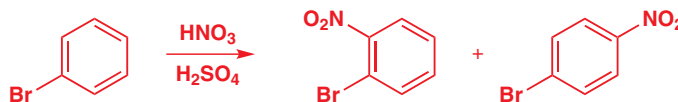


If the reaction is irreversible, the α -product is usually formed. But if the reaction is reversible, as is the case with sulfonation, the position of substitution may be determined by temperature. Sulfonation at low temperatures gives the α -product by kinetic control, while sulfonation at high temperatures gives the β -product by thermodynamic control. The β -product is formed more slowly but it is more stable as there is less steric hindrance between the large sulfonic acid group and the orange hydrogen on the other ring. Under conditions allowing reversible sulfonation, eventually all the product ends up β .

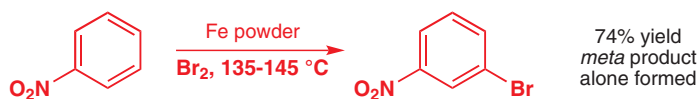


Regiocontrol by choice of route

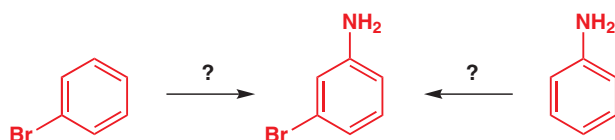
Choosing the right route to an aromatic product is essential if you want to get one particular isomer. We can illustrate this with the synthesis of the isomers of bromonitrobenzene. Because the bromo substituent is *ortho*, *para*-directing and the nitro group *meta*-directing, it's possible to make all three isomers, providing we exploit the regioselectivity of electrophilic substitution. Nitration of bromobenzene would give the *ortho* and *para* isomers while bromination of nitrobenzene would give the *meta* isomer. The selectivity of the first reaction is not good: bromine is small and not very electronegative, so steric hindrance is weak and the *ortho* positions are not deactivated. Furthermore there are two *ortho* positions but only one *para*: a typical result is about 37% *ortho*, 1% *meta*, and 62% *para*. Both compounds are industrial products, made by nitration and separated.



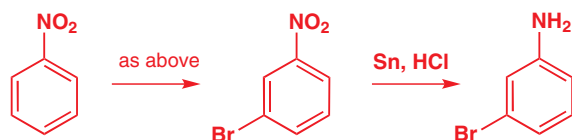
Bromination of nitrobenzene is remarkably good, considering the unreactivity of nitrobenzene in electrophilic aromatic substitution. One recipe uses iron powder and bromine at 140 °C and gives 74% of the *meta* product. We shall need these reactions in the next section.



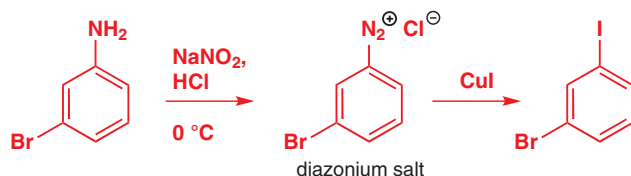
Before we move on, consider why this selectivity works: we can get all three isomers because we have one *ortho/para* director and one *meta* director. But what if we had two *ortho/para* directors—say, amino and bromo—and wanted the *meta* isomer?



The solution in these cases is often to make use of the transformation of the nitro group (a *meta* director) into an amino group (a *para* director) by reduction.



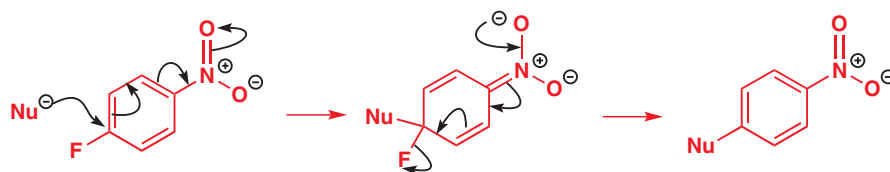
Since the amino group can be substituted by diazotization (p. 520), many problems of regioselectivity can be solved by using nitro compounds as intermediates. You could, for example, use the product above to make the otherwise challenging 3-bromoiodobenzene:



➡ Diazonium salts, and their use in the synthesis of aromatic compounds, were discussed on pp. 495 and 520.

Regioselectivity in nucleophilic aromatic substitution

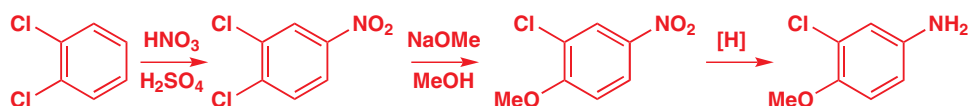
As you saw in Chapters 21 and 22, diazonium salts need no activation to undergo nucleophilic aromatic substitution, but for other leaving groups a nitro group is commonly used as an activator. The three fluoronitrobenzenes are all commercial products but only the *ortho* and *para* isomers can do the nucleophilic substitution. This is because the nitro group must be able to stabilize the addition intermediate by accepting the negative charge.



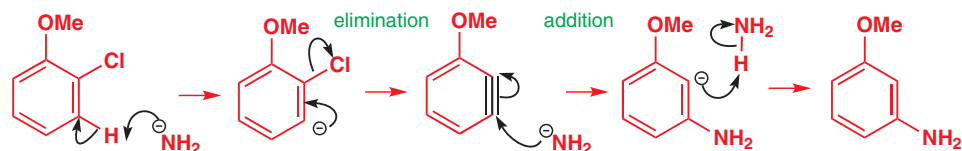
By carefully combining electrophilic and nucleophilic substitution it is possible to make aromatic compounds with substituents arranged in a precise and predictable fashion. So, if we nitrate *o*-dichlorobenzene, all positions are favourable but the nitro group goes in *para* to one Cl atom because of steric hindrance at the *ortho* positions. Although chlorine is small, two chlorines next to each other have a buttressing effect as each pushes the other away. It is difficult to get three adjacent substituents on a benzene ring. If we now do a nucleophilic

➡ The various ways to carry out nucleophilic aromatic substitution are described on pp. 514–526.

aromatic substitution, only the Cl *para* to the nitro group is displaced. We can even reduce the nitro group to the corresponding amine.

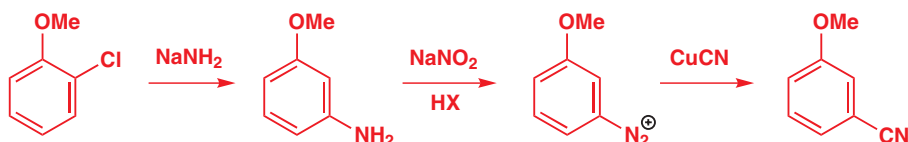


The last successful method for nucleophilic aromatic substitution uses a benzyne intermediate—on p. 524 you saw benzyne chemistry being used to make *meta*-aminoanisole, like this:



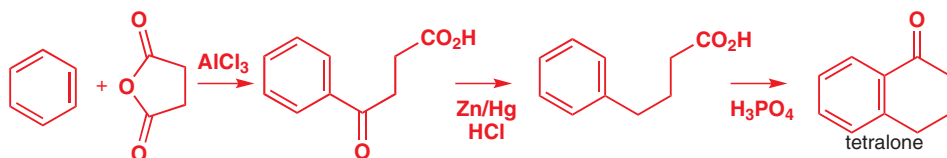
➡ The regiochemistry of this reaction is explained in Chapter 22.

Now that the amino group is fixed; we can displace it via a diazonium salt using any chosen nucleophile—copper cyanide for example:



Regioselectivity of intramolecular reactions

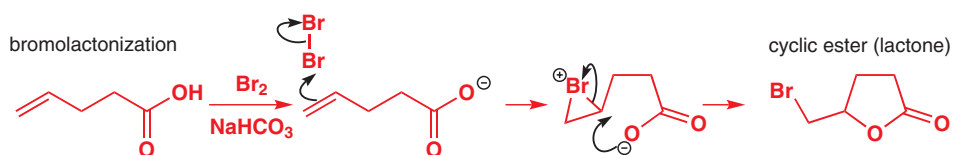
A cunning way to get unusual regioselectivity is to make the reaction intramolecular. The synthesis from benzene of the cyclic ketone known as tetralone may look difficult as we must get an *ortho* relationship on the benzene ring. But if we make the final bond in the ring by a Friedel–Crafts acylation there is no problem. The alkyl group is *ortho,para*-directing and the acid cannot reach the *para* position.



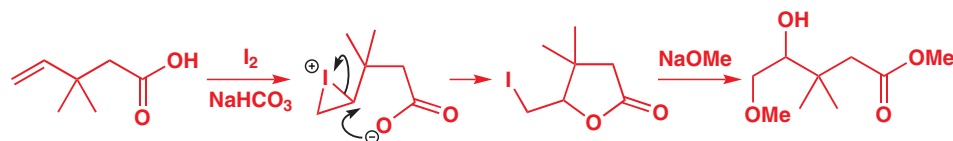
■ Usually a more powerful catalyst (AlCl_3) is needed, but intramolecular acylations are fast enough without this.

Notice the use of a cyclic anhydride in the first Friedel–Crafts acylation. It doesn't matter where the acylation occurs and the reaction stops there as the ring is deactivated by the ketone and the carboxylic acid released in the reaction is much less electrophilic than the anhydride. The ketone is then reduced to a CH_2 group by the Clemmensen method (see Chapter 23) and polyphosphoric acid is used to carry out the intramolecular acylation step.

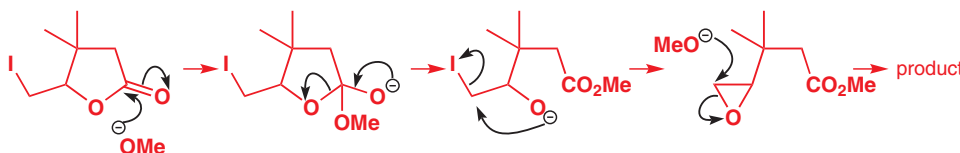
A more subtle approach is to use a 'tether'—something that holds two reagents together and is afterwards cleaved. An example is halolactonization. The idea is simple. A halogen, say bromine, attacks an alkene and the bromonium ion intermediate is captured intramolecularly by the anion of a carboxylic acid. The reaction therefore uses bromine and NaHCO_3 —a weak base, but one strong enough to deprotonate a carboxylic acid. The anion attacks the more highly substituted end of the bromonium ion, as explained in Chapter 19, and forms a five-membered ring.



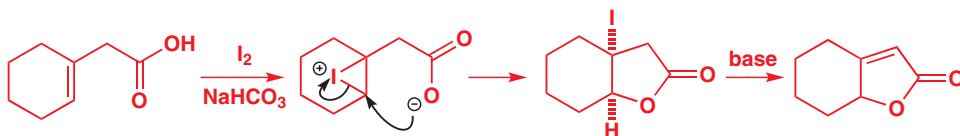
Although any halogen might be used in this reaction, iodine is the most versatile and the reaction is commonly called iodolactonization. The tether is the C–O bond of the lactone and this can be cleaved with an alkoxide.



The reaction with methoxide needs some explanation. Attack on the carbonyl group cleaves the lactone, releasing an alkoxide that cyclizes to form an epoxide. A second molecule of methoxide now attacks the epoxide, opening it from the less hindered end as we should expect in an anionic reaction (Chapter 19).



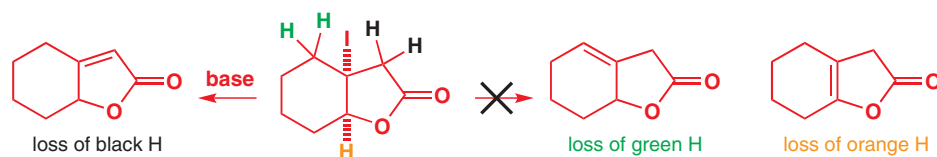
Another example shows that reaction may occur at the other end of the iodonium ion. Attack at the tertiary carbon would be difficult sterically and, in any case, would give an unstable four-membered ring. The lactone formed has the iodine β to the carbonyl group and so eliminates easily in base (pyridine works well) by the E1cB mechanism (Chapter 17) to give the unsaturated lactone. Although the relative stereochemistry of the iodolactone is controlled by the inversion in the opening of the iodonium ion, it is irrelevant as it disappears in the elimination step.



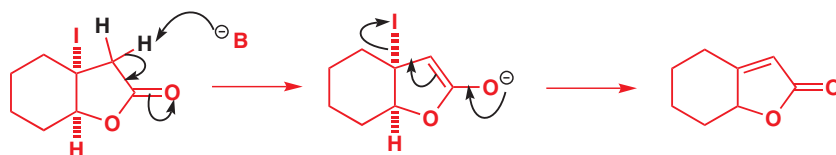
➡ We'll come back to the use of iodolactonization to control stereochemistry in Chapter 32.

Regioselectivity in elimination reactions

This question was discussed in Chapter 17 but we can return to it here with more sophisticated examples. The regioselectivity in the last reaction of the sequence above dictates the position of the alkene in the product. Of all of the protons adjacent to the iodo group, only a black one is lost:



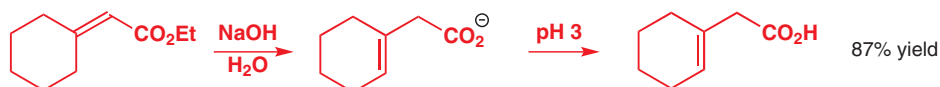
The orange hydrogen cannot be lost by E2 as it is *cis* to the iodine and E2 reactions prefer a *trans* (anti-periplanar) arrangement. The green hydrogens are not lost because they are less acidic than the black hydrogens. In fact, this is not an E2 elimination at all. Because one of the black hydrogens can be lost in enolate formation, this is an E1cB elimination.



But now another regioselectivity question arises: if elimination occurs preferably towards the carbonyl group, how can we make the starting material for the iodolactonization sequence, which has the alkene not in conjugation with the carboxylic acid? It turns out that it is better to make the ester with the ‘wrong’ regioselectivity. This is easily done by a Horner–Wadsworth–Emmons reaction using a phosphonate ester. This Wittig-style reaction is explained in Chapter 27.



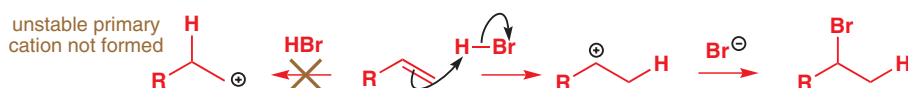
Now comes the remarkable regioselectivity. The ester is hydrolysed, as usual, in aqueous NaOH. On acidification to pH 3, the free acid is released and the double bond has moved into the ring.



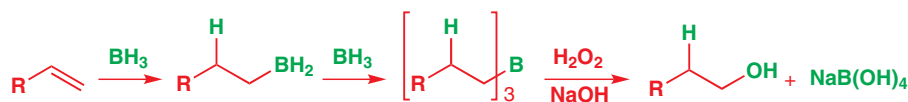
Alkenes like to be conjugated with carbonyl groups but they also prefer to be *inside* six-membered rings rather than outside—in this case presumably because the ester group otherwise has to eclipse a ring carbon. Conjugation with an ester group pulls the alkene out of the six-membered ring in the lactone we made above, but when the carbonyl group is a carboxylate anion, conjugation is very weak and the double bond moves into the ring.

Electrophilic attack on alkenes

You met electrophilic attack on alkenes in Chapter 19 and we shall just briefly revisit its regioselectivity. Unsymmetrical alkenes add HBr to give the more stable of the two possible cations. If R is alkyl or aryl, this means the more substituted cation.

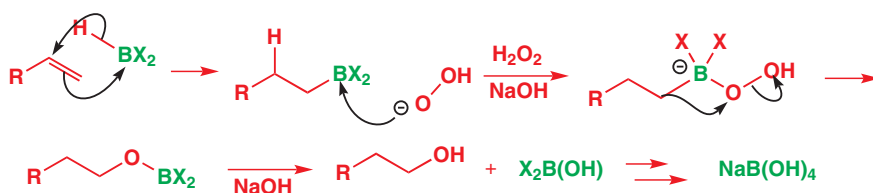


If you want to get the other regioisomer, with the heteroatom at the end, you can use hydroboration (Chapter 19) or the radical reactions described in the next section. Here is a brief reminder of hydroboration. Reaction between a borane having at least one B–H bond with an alkene gives an alkyl borane in which all the hydrogens are replaced by alkyl groups. Oxidation gives the terminal alcohol.



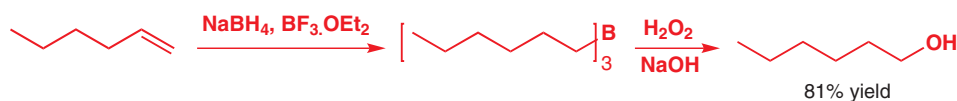
The regioselectivity comes from the first step. The boron's empty p orbital bonds to the more nucleophilic end of the alkene and hydride is transferred to give a borane. Reaction with alkaline H₂O₂ leads to migration of an alkyl group from boron to oxygen and eventually to the alcohol.

■ In these structures X can be R or H.



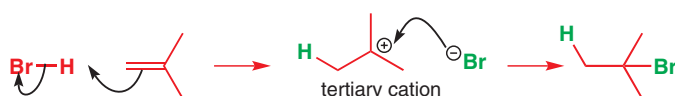
Interactive mechanism for hydroboration

Borane is unstable but can easily be made from NaBH_4 and BF_3 . In this synthesis of hexan-1-ol from hex-1-ene, a water molecule has been added to the alkene, but with the opposite regioselectivity to reactions with H_2O in acid or HBr .



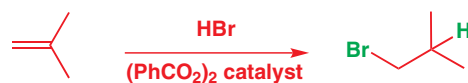
Regioselectivity in radical reactions

Almost every reaction we have discussed so far has been ionic, but in this short section we need to give you a preview of another group of reactions we return to in Chapter 37—those of *radicals*. When HBr adds to an unsymmetrical alkene we use arrows that represent the movement of two electrons to give charged intermediates that combine in a second step to give a neutral product. The strong $\text{H}-\text{Br}$ bond breaks to give a bromide ion and a stable alkyl cation. This bond breaks *heterolytically*—that is, unsymmetrically—as does the alkene bond. We can predict the regiochemistry of these reactions by making the most stable anions and cations as intermediates, in this case a tertiary alkyl cation and a bromide anion.

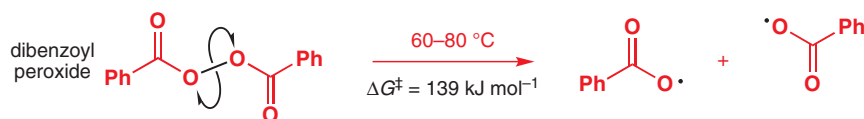


Radical addition

The regioselectivity in the reaction below is opposite: a primary alkyl bromide is formed, by a different mechanism involving radicals.



In radical reactions, bonds break *homolytically* with one electron going one way and one the other. The radicals that are formed have an odd number of electrons, one of which must be unpaired. This makes them very reactive and they are not usually isolated. Even strong bonds can break into ions provided they are polarized, but to make radicals we need weak symmetrical bonds such as $\text{O}-\text{O}$, $\text{Br}-\text{Br}$ or $\text{I}-\text{I}$. Dibenzoyl peroxide, the $\text{Ph}(\text{CO}_2)_2$ catalyst in this reaction, readily undergoes homolysis like this—the one-electron movements are represented by ‘fish-hook’ arrows having one barb and odd electrons on atoms are represented by dots.



Now we can use the new radicals we have just made to cleave the strong HBr bond homolytically because a new and very strong OH bond will be formed. As we start with one radical intermediate that must have an unpaired electron, we must finish with another radical with an unpaired electron. In this case, it is a bromine radical.



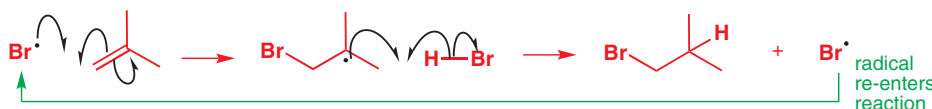
If we do this reaction in the presence of the alkene we have just reacted with HBr , the bromine radical adds to the alkene in one of the two possible ways. Although radicals are neutral,

➡ You'll meet radicals in much greater detail in Chapter 37.

they are electron-deficient (the C atom is one electron short) and, rather like cations, are more stable the more substituents they have. So the tertiary radical is formed rather than the primary radical, and the bromine ends up at the primary position.

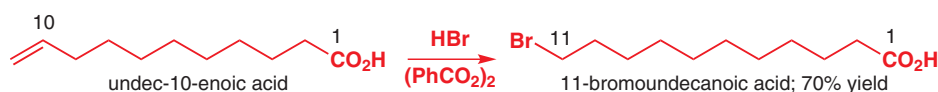


We have still not reached the end of the reaction as our product is still a radical. How can it become a molecule with only paired electrons? The answer is simple. It reacts with another HBr molecule to produce more bromine radicals. Now you see something important to all radical reactions: only a small amount of the radical is needed as more radicals are produced every time the reaction gives product. The overall process is a *radical chain reaction*.

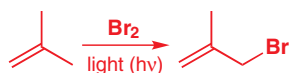


Interactive mechanism for radical addition of HBr to alkenes

Because of this we also need only very small amounts of dibenzoyl peroxide, the radical initiator, which is just as well as it is potentially explosive, like many radical generators. Here is the reaction being used to make a bromoacid:



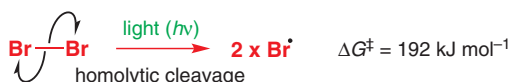
Radical abstraction



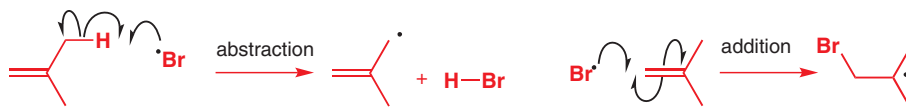
We sneaked a new reaction into that sequence. The removal of a hydrogen atom (note: not a proton) from HBr by the peroxide radical is an *abstraction* reaction. The bromine radical will also abstract hydrogen atoms and will do so from the same alkene we have just used but with yet another different outcome, as you see in the margin.

When light shines on bromine, the weak Br–Br bond breaks to give two bromine radicals. Heat will do the job too but light is cleaner and, as bromine is brown, it absorbs most wavelengths of visible light.

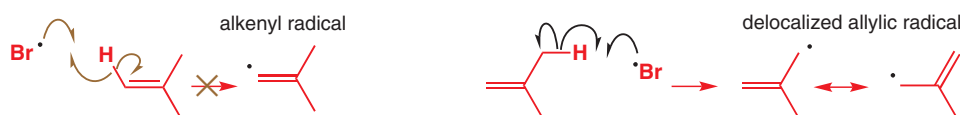
■ Note that the Br–Br bond is more stable than the O–O bond in the peroxide.



Radicals are very unstable and reactive, and these bromine radicals may simply recombine or they may react with other compounds. You already know that bromide anions are good nucleophiles in S_N2 reactions, but bromine radicals do two quite different reactions: abstraction and addition. The Br radical may abstract a hydrogen atom from the alkene or it may add to the π bond. Notice that each reaction produces a new carbon-centred radical and, in the first case, a molecule of HBr. Whereas the Br–Br bond is weak, the H–Br bond is much stronger (366 kJ mol^{-1}) and, unlike ionic reactions, radical reactions are dominated by bond strength.

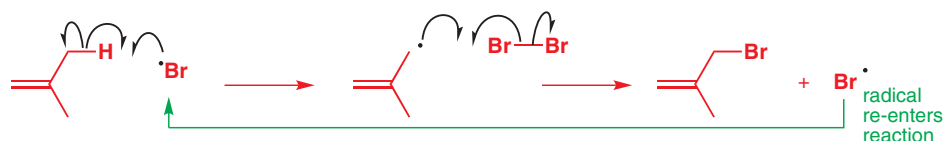


The first reaction introduces another important aspect of regioselectivity: why does the radical abstract that H atom, and not one from the alkene?



Removal of an alkene H gives a carbon-centred radical localized on the sp^2 atom but the removal of an H from a methyl group gives a much more stable delocalized allylic radical. In addition there are six such H atoms but only two alkene H atoms.

The reaction obviously cannot end there with the formation of another radical, however stable, and this allylic radical collects a bromine atom from a bromine molecule. Note that the allylic radical doesn't react with a bromine *radical* in this step: radicals are very unstable and the concentration of radicals at any one time is so low that it is rare for two of them to meet.

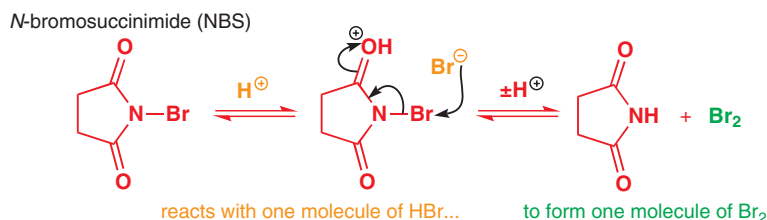


 Interactive mechanism for allylic bromination

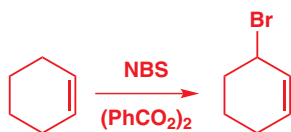
This step also produces a new bromine radical that can start a new series of reactions. Like the addition of HBr above, the reaction is a radical chain reaction, and only a small amount of Br_2 needs to break down to $Br^\bullet$ to get the reaction going. This is important as you already know what happens when bromine molecules react with alkenes: addition occurs by an ionic mechanism. Add too much Br_2 and the bromine molecules attack the alkene directly and do not abstract H atoms.



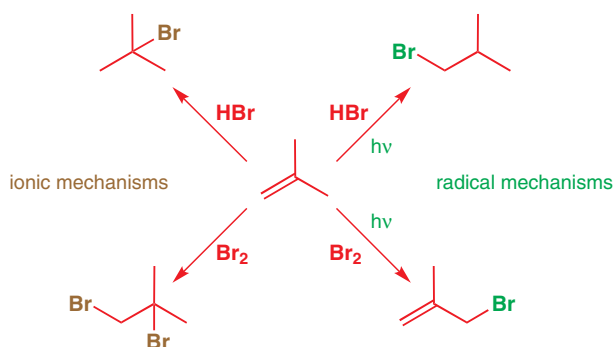
If we want to make the dibromide, we use plenty of bromine, but if we want to use a radical process to make the allylic bromide we must take advantage of the greater reactivity of the radical and keep the bromine concentration low. A good way to do this is to use the compound NBS (*N*-bromosuccinimide), which you met in Chapter 19. NBS acts as a sort of turnstile which only lets a molecule of Br_2 out when a molecule of HBr is formed (and of course HBr is the by-product in the radical bromination).



Br_2 is slowly released into the reaction as it proceeds, and the concentration never builds up enough to generate the dibromide. In this example, dibenzoyl peroxide is the initiator and allylic bromination gives the useful cyclohexenyl bromide.



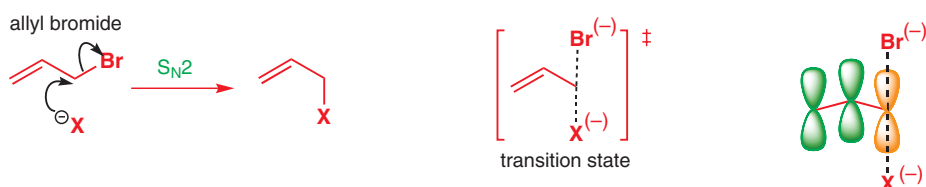
These radical reactions will be described in much greater detail in Chapter 37. For the moment you need only notice that they can have quite different regioselectivity from ionic reactions with the same reagents.



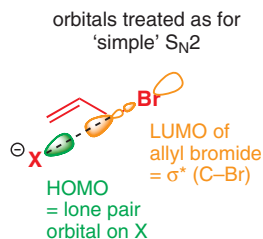
Nucleophilic attack on allylic compounds

The allylic bromides that can be made by these radical reactions display interesting regioselectivity. We shall start with some substitution reactions with which you are familiar from Chapter 15. There we said that allyl bromide is about 100 times more reactive towards simple S_N2 reactions than is propyl bromide or other saturated alkyl halides.

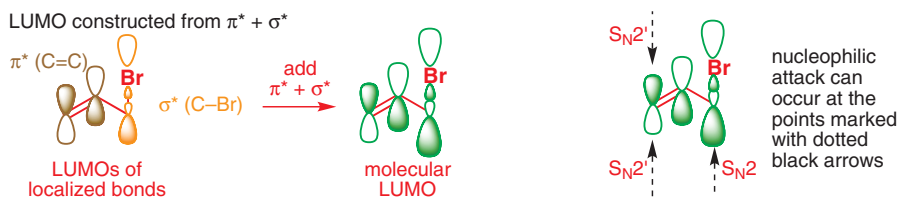
The double bond stabilizes the S_N2 transition state by conjugation with the p orbital at the carbon atom under attack. This full p orbital (shown in orange in the diagram below) forms a partial bond with the nucleophile and with the leaving group in the transition state. Any stabilization of the transition state will, of course, accelerate the reaction by lowering the energy barrier.



There is an alternative mechanism for this reaction that involves nucleophilic attack on the alkene instead of on the saturated carbon atom. This mechanism leads to the same product and is often called the S_N2' (pronounced 'S-N-two-prime') mechanism.

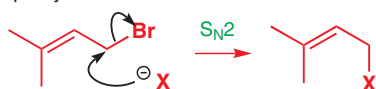


We can explain both mechanisms in a unified way if we look at the frontier orbitals involved. The nucleophile must attack an empty orbital (the LUMO), which we might expect to be simply σ* (C-Br) for the S_N2 reaction. But this ignores the alkene. The interaction between π* (C=C) and the adjacent σ* (C-Br) will as usual produce two new orbitals, one higher and one lower in energy. The lower-energy orbital, π* + σ*, will now be the LUMO. To construct this orbital we must put all the atomic orbitals parallel and make the contact between π* + σ* a bonding interaction.

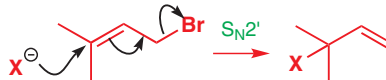


If the allylic halide is unsymmetrically substituted, a question of regioselectivity arises. The products from S_N2 and S_N2' are different and the normal result is that nucleophilic attack occurs at the less hindered end of the allylic system, whether that means S_N2 or S_N2' . This important allylic bromide, known as prenyl bromide, normally reacts entirely via the S_N2 reaction.

prenyl bromide reacts like this

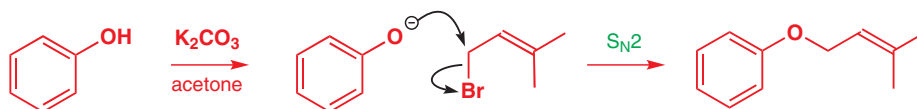


and not like this

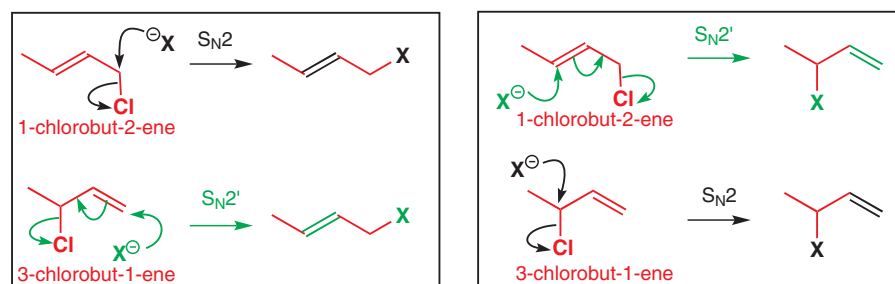


The two ends of the allylic system are contrasted sterically: direct (S_N2) attack is at a primary carbon while allylic (S_N2') attack is at a tertiary carbon atom so that steric hindrance favours the S_N2 reaction. In addition, the number of substituents on the alkene product means that the S_N2 product is nearly always preferred— S_N2 gives a trisubstituted alkene while the S_N2' product has a less stable monosubstituted alkene.

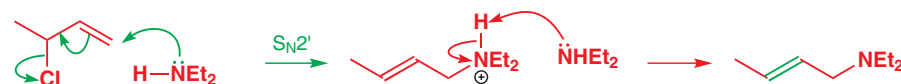
An important example is the reaction of prenyl bromide with phenols. This is simply carried out with K_2CO_3 in acetone as phenols are acidic enough ($pK_a \sim 10$) to be substantially deprotonated by carbonate. The product is almost entirely from the S_N2 route, and is used in the Claisen rearrangement (Chapter 35).



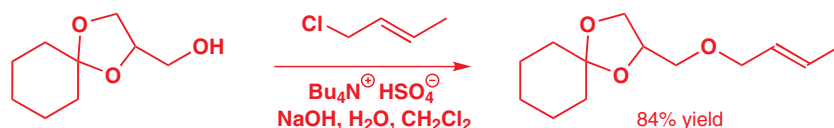
If we make the two ends of the allyl system more similar, say one end primary and one end secondary, things are more equal. We could consider the two isomeric butenyl chlorides.



All routes look reasonable, although we might again expect faster attack at the primary carbon. The reactions in the left-hand box are preferred to those in the right-hand box. But there is no special preference for the S_N2 over the S_N2' mechanism or vice versa—the individual case decides. If we react the secondary butenyl chloride with an amine we get the S_N2' mechanism entirely.



If the primary chloride is used, once again we get nucleophilic attack at the primary centre. The more stable product with the more highly substituted alkene is formed this time by the S_N2 reaction. Here is a slightly more advanced example:



■ So far we have used the word 'allyl' to describe these compounds. Strictly, that word applies only to specific compounds $CH_2=CH-CH_2X$ with no substituents other than hydrogen. Allyl is often used loosely to describe any compound with a functional group on the carbon atom *next* to the alkene. We shall use 'allylic' for that and 'allyl' only for the unsubstituted version.

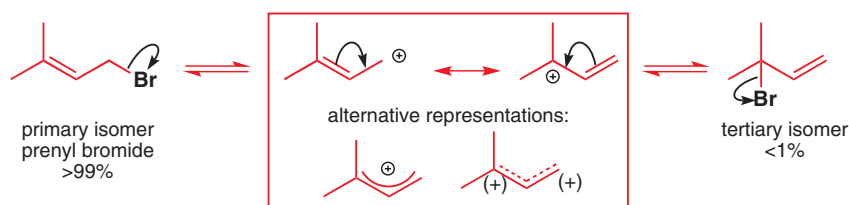
Interactive mechanisms for various nucleophilic substitutions

Interactive mechanism for S_N2' nucleophilic substitutions

➡ We explained why adjacent double bonds assist S_N2 reactions on p. 341.

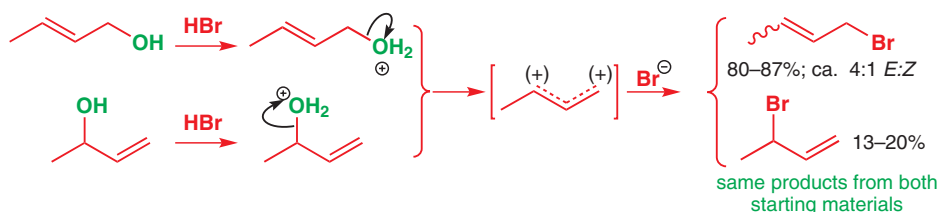
Notice that these reactions take place with allylic *chlorides*. We should not expect an alkyl chloride to be particularly good at S_N2 reactions as chloride ion is only a moderate leaving group and we should normally prefer to use alkyl bromides or iodides. *Allylic* chlorides are more reactive because of the alkene. Even if the reaction occurs by a simple S_N2 mechanism without rearrangement, the alkene is still making the molecule more electrophilic.

You might ask a very good question at this point. How do we know that these reactions really take place by S_N2 and S_N2' mechanisms and not by an S_N1 mechanism via the stable allyl cation? Well in the case of prenyl bromide, we don't! In fact, we suspect that the cation probably *is* an intermediate because prenyl bromide and its allylic isomer are in rapid equilibrium in solution at room temperature.

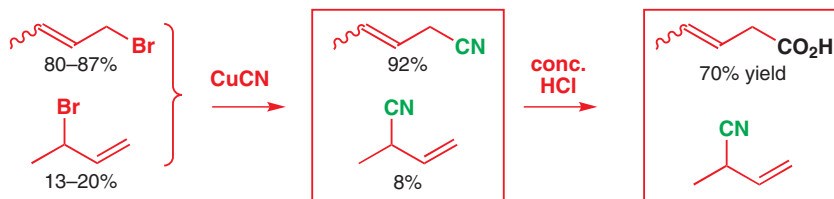


The equilibrium is entirely in favour of prenyl bromide because of its more highly substituted double bond. Reactions on the tertiary allylic isomer are very likely to take place by the S_N1 mechanism: the cation is stable because it is tertiary and allylic and the equilibration tells us it is already there. Even if the reactions were bimolecular, no S_N2' mechanism would be necessary for the tertiary bromide because it can equilibrate to the primary isomer more rapidly than the S_N2 or S_N2' reaction takes place.

Even the secondary system we also considered is in rapid equilibrium when the leaving group is bromide. This time both allylic isomers are present, and the primary allylic isomer (known as crotyl bromide) is an *E/Z* mixture. The bromides can be made from either alcohol with HBr and the same ratio of products results, indicating a common intermediate in the two mechanisms. You saw at the beginning of Chapter 15 that this reaction is restricted to alcohols that can react by S_N1 .



Displacement of the bromide by cyanide ion, using the copper(I) salt as the reagent, gives a mixture of nitriles in which the more stable primary nitrile predominates even more. These can be separated by a clever device. Hydrolysis in concentrated HCl is successful with the predominant primary nitrile but the more hindered secondary nitrile does not hydrolyse. Separation of compounds having two different functional groups is easy: in this case the acid can be extracted into aqueous base, leaving the neutral nitrile in the organic layer.

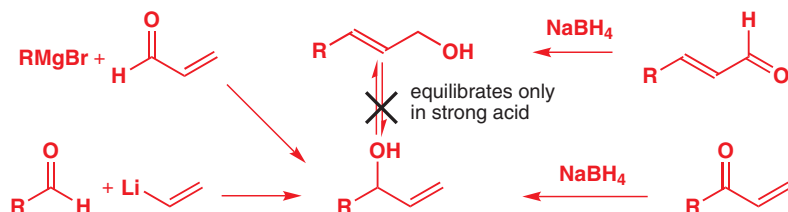


Once again, we do not know for sure whether this displacement by cyanide goes by the S_N1 or S_N2' mechanism, as the reagents equilibrate under the reaction conditions. However, the

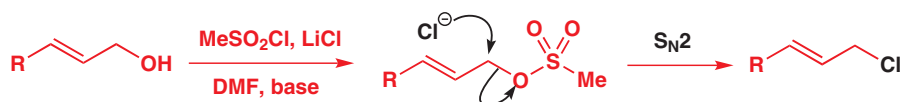
chlorides do *not* equilibrate and so, if we want a clear-cut result on a single well-defined starting material, the chlorides are the compounds to use. But you already see that regioselectivity with allylic compounds may depend on steric hindrance, rates of reaction, and stability of the product.

Regiospecific preparation of allylic chlorides

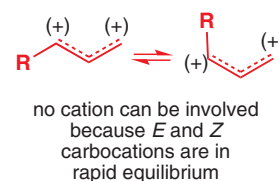
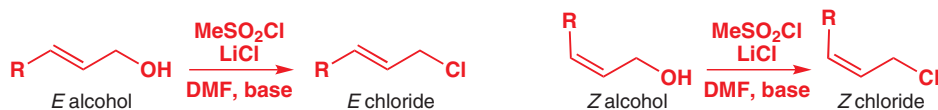
Allylic alcohols are good starting materials for making allylic compounds with control over where the double bond and the leaving group will be. Allylic alcohols are easily made by addition of Grignard reagents or organolithium compounds to enals or enones (Chapter 9) or by reduction of enals or enones (Chapter 23). More to the point, they do not equilibrate except in strongly acidic solution, so we know which allylic isomer we have.



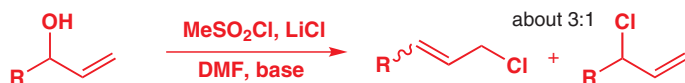
Conversion of the alcohols into the chlorides is easier with the primary than with the secondary alcohols. We need to convert OH into a leaving group and provide a source of chloride ion to act as a nucleophile. One way to do this is with methanesulfonyl chloride (MeSO_2Cl) and LiCl .



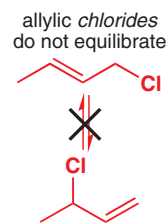
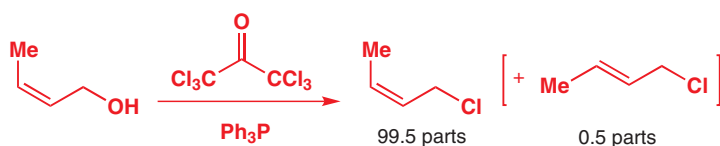
This result hardly looks worth reporting and, anyway, how do we know that equilibration or $\text{S}_{\text{N}}1$ reactions aren't happening? Well, here the mechanism must be $\text{S}_{\text{N}}2$ because the corresponding *Z*-allylic alcohol preserves its alkene configuration. If there were equilibration of any sort, the *Z*-alkene would give the *E*-alkene because *E*- and *Z*-allylic cations are not geometrically stable.



Sadly, this method fails to preserve the integrity of the secondary allylic alcohol, which gives a mixture of allylic chlorides.



Reliable clean $\text{S}_{\text{N}}2$ reactions with secondary allylic alcohols can be achieved only with Mitsunobu chemistry. Here is a well-behaved example with a *Z*-alkene. The reagents have changed since your last encounter with a Mitsunobu-type reaction: instead of DEAD and a carboxylic acid we have hexachloroacetone, with, of course, triphenylphosphine.

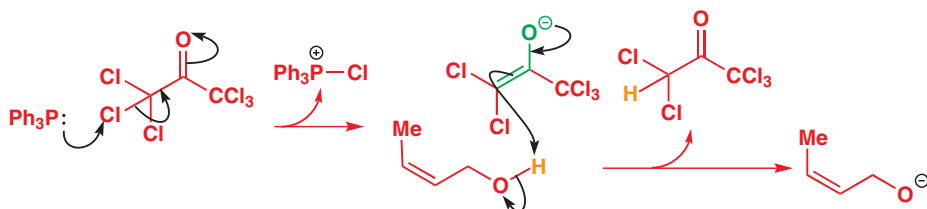


By analogy with *stereospecific*, we can define *regiospecific* to mean a reaction where the regiochemistry (that is, the location of the functional groups) of the product is determined by the regiochemistry of the starting material.

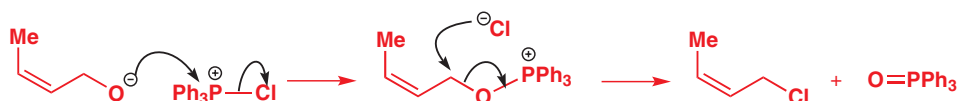
➡ The Mitsunobu reaction was discussed in Chapter 15, p. 349. Mitsunobu chemistry involves using a phosphorus atom to remove the OH group, after the style of PBr_3 as a reagent to make alkyl bromides from alcohols.

The first thing that happens is that the lone pair on phosphorus attacks one of the chlorine atoms in the chloroketone. The leaving group in this S_N2 reaction at chlorine is an enolate, which is a basic species and can remove the proton from the OH group in the allylic alcohol.

■ Phosphorus doing a substitution at a C–Cl bond the wrong way round? But P is soft, so it cares little about the polarization of the bond, only about the energy of the C–Cl σ^* . The energy is the same whichever end of the bond is attacked. You may see similar reactions of PPh_3 with CBr_4 or CCl_4 : all produce stabilized carbanions.



Now the alkoxide anion can attack the positively charged phosphorus atom. This is a good reaction in two ways. First, there is the obvious neutralization of charge and, second, the P–O bond is very strong.

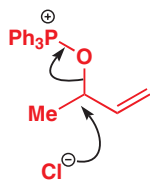


■ We looked at the converse—'loose' S_N2 transition states with considerable S_N1 character—in the reactions of bromonium ions and protonated epoxides in Chapter 17.

The next step is a true S_N2 reaction at carbon as the very good leaving group is displaced. The already strong P–O single bond becomes an even stronger P=O double bond to compensate for the loss of the strong C–O single bond. There is obviously no S_N1 component in this displacement (otherwise the *Z*-alkene would have partly isomerized to the *E*-alkene) and very little S_N2' presumably as only 0.5% of the rearrangement product is formed. These displacements of $Ph_3P=O$ are often the 'tightest' of S_N2 reactions.

Now for the really impressive result. Even if the alcohol is secondary, and the rearranged product would be thermodynamically more stable, very little of it is formed and almost all the reaction is clean S_N2 .

S_N2 preferred to S_N2'

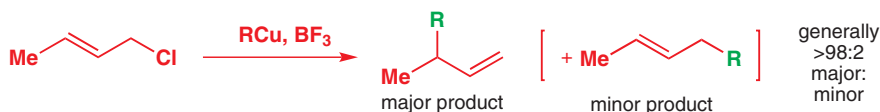


There is a bit more rearrangement than there was with the other isomer but that is only to be expected. The very high proportion of direct S_N2 product shows that there is a real preference for the S_N2 over the S_N2' reaction in this displacement.

Now that we know how to make allylic chlorides of known structure—whether primary or secondary—we need to discover how to replace the chlorine with a nucleophile with predictable regioselectivity. We have said little so far about carbon nucleophiles (except cyanide ion) so we shall concentrate on simple carbon nucleophiles in the S_N2' reaction of allylic chlorides.

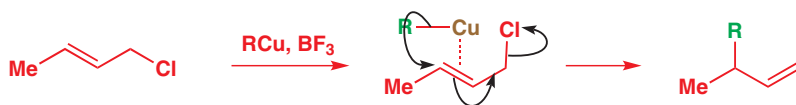
The S_N2' reaction of carbon nucleophiles on allylic chlorides

Ordinary carbon nucleophiles such as cyanide or Grignard reagents or organolithium compounds fit the patterns we have described already. They usually give the more stable product by S_N2 or S_N2' reactions depending on the starting material. If we use copper compounds, there is a tendency—no more than that—to favour the S_N2' reaction. You will recall that copper(I) was the metal we used to ensure conjugate addition to enones (Chapter 22) and its use in S_N2' reactions is obviously related. Simple alkyl copper reagents (RCu , known as Gilman reagents) generally favour the S_N2' reaction but we can do much better by using RCu complexed with BF_3 .

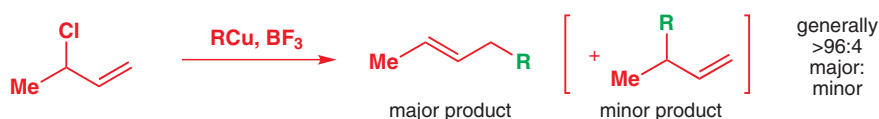


➡ The nature of metal–alkene complexes is discussed in Chapter 40.

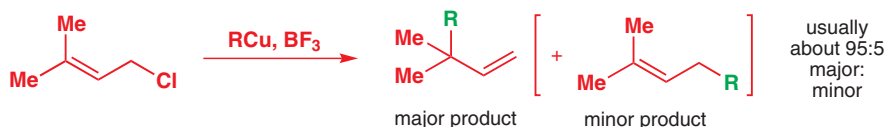
The copper must complex to the alkene and then transfer the alkyl group to the S_N2' position as it gathers in the chloride. This might well be the mechanism, although it is often difficult to draw precise mechanisms for organometallic reactions.



The secondary allylic isomer also gives almost entirely the rearranged product. This is perhaps less surprising, as the major product is the more stable isomer, but it means that either product can be formed in high yield simply by choosing the right (or should we say *wrong*, since there is complete allylic rearrangement during the reaction) isomer. The reaction is *regiospecific*.



The most remarkable result of all is that prenyl chloride gives rearranged products in good yield. This is about the only way in which these compounds suffer attack at the tertiary centre by S_N2' reaction when there is the alternative of an S_N2 reaction at a primary centre.



Electrophilic attack on conjugated dienes

Another way to make allylic chlorides is by treating dienes with HCl. Electrophiles attack conjugated dienes more readily than they do isolated alkenes. There was some discussion of this in Chapter 19, establishing the main point that the terminal carbon atoms are the most nucleophilic and that the initial attack produces an allylic cation. A simple example is the addition of HCl to cyclopentadiene.

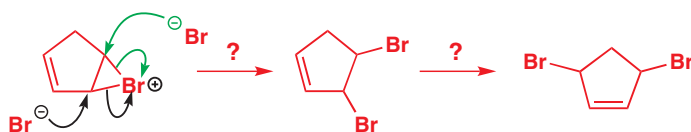


Although there is a question of regioselectivity in the initial protonation, the allylic cation is symmetrical and attack by chloride at either end produces the same product. However, if the electrophile is a halogen rather than HCl or HBr then the reaction becomes regioselective as the cationic intermediate is no longer symmetrical. What happens is this:



The alternative is direct attack on the bromonium ion intermediate, which we assume would occur at the allylic site (black arrows) and not at the other (green arrows). Although this 1,2-dibromide product is not observed, it is still possible that this reaction happens because the 1,2-product can rearrange by bromide shift to the observed 1,4-dibromide.

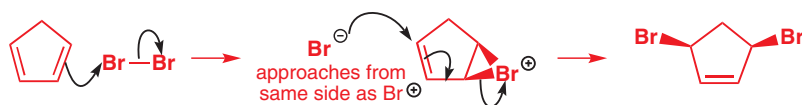
➡ By 'bromide shift' we mean the reversible isomerization of allylic bromides you saw on p. 576.



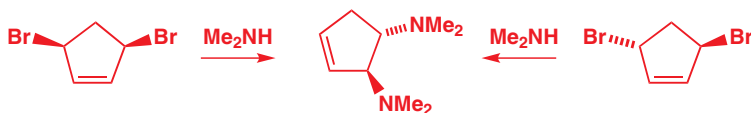
The final product of this reaction could in fact be either of two compounds as the two bromine atoms may be *cis* or *trans*. Bromination in chloroform at $-20\text{ }^{\circ}\text{C}$ gives mostly a liquid *cis* dibromide while reaction in hydrocarbon solvents gives the crystalline *trans* isomer. On standing the *cis* isomer slowly turns into the *trans*.



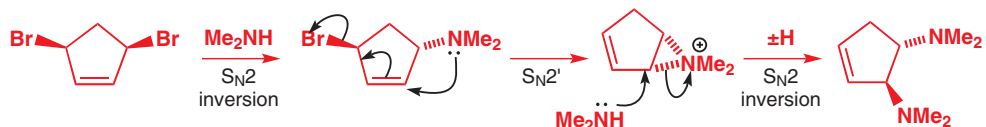
This suggests that the *cis* bromide is the kinetic product and the more stable *trans* compound is the thermodynamic product, formed by reversible loss of bromide and reformation of the bromonium ion.



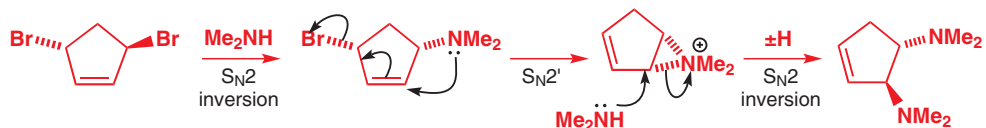
Similar questions arise when nucleophilic substitution occurs on the dibromides. Reaction of either the *cis* or the *trans* dibromide with dimethylamine gives the *trans* isomer of a diamine. But look at the regioselectivity—it's not the diamine you might expect. The only explanation is one $\text{S}_{\text{N}}2$ displacement and one $\text{S}_{\text{N}}2'$ displacement.



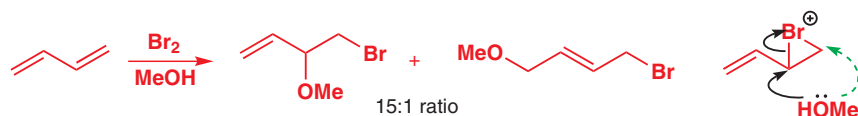
But what about the stereochemistry? Starting with the *cis* isomer, one $\text{S}_{\text{N}}2$ displacement with inversion might be followed by an intramolecular $\text{S}_{\text{N}}2'$ displacement and finally another $\text{S}_{\text{N}}2$ displacement with inversion at the allylic centre.



The reaction with the *trans* isomer is almost identical: the same three-membered ring is an intermediate in both sequences so the products are bound to be the same.

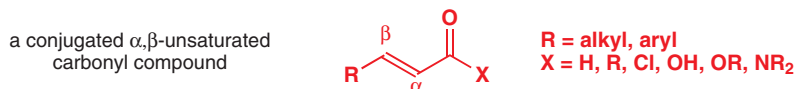


If the nucleophile is different from the electrophile we can get a bit more information about the course of the reaction. When butadiene is treated with bromine in methanol as solvent, two adducts are formed in a 15:1 ratio along with some dibromide. Methanol is a weak nucleophile and adds to the bromonium ion mainly at the allylic position (black arrow below); only a small amount of product is formed by attack at the far end of the allylic system. Note that no attack occurs at the other end of the bromonium ion (green dotted arrow).

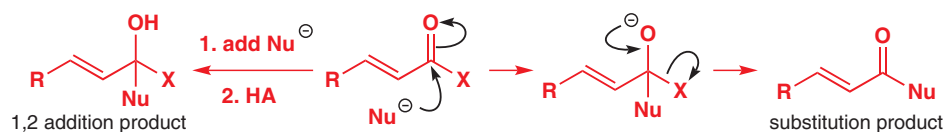


Conjugate addition

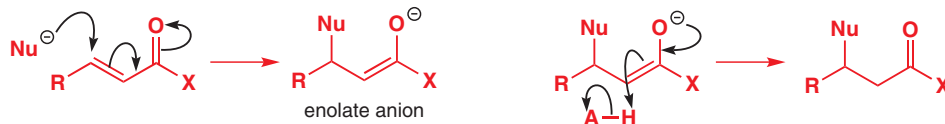
In Chapter 22 we devoted considerable space to discussing conjugate addition and the reasons why some reactions occur by direct attack on the carbonyl group of an α,β -unsaturated carbonyl compound and why others occur by conjugate addition. We shall briefly revise the regioselectivity aspects of these reactions.



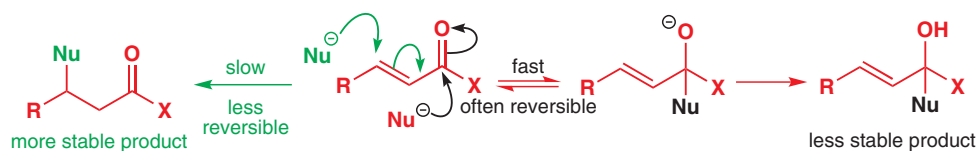
Direct (or 1,2) addition means that the nucleophile attacks the carbonyl group directly. An addition compound is formed which may lose X^- , if it is a leaving group, or become protonated to give an alcohol.



Conjugate (or 1,4) addition means that the nucleophile adds to the end of the alkene furthest from the carbonyl group. The electrons move through into the carbonyl group to produce an enolate anion that usually becomes protonated to give a ketone.



The first difference between the two routes is that the product from direct addition keeps the alkene but loses the carbonyl group while conjugate addition keeps the carbonyl group but loses the alkene. As a $C=O$ π bond is stronger than a $C=C$ π bond, **conjugate addition gives the thermodynamic product**. But as the carbonyl group is more electrophilic than the far end of the alkene, especially to charged, hard nucleophiles, **direct addition gives the kinetic product**. So direct addition is favoured by low temperatures and short reaction times while conjugate addition is favoured by higher temperatures and longer reaction times, provided the 1,2 addition is reversible.



The second difference depends on how electrophilic is the α,β -unsaturated carbonyl compound. The more electrophilic such as aldehydes and acid chlorides tend to prefer direct addition while the less electrophilic such as ketones or esters tend to prefer conjugate addition.



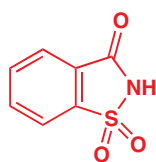
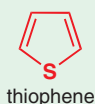
It is similar with the choice of nucleophile: more nucleophilic species, such as MeLi or Grignard reagents, prefer direct addition, particularly as they react irreversibly, while less nucleophilic species like amines and thiols prefer conjugate addition. These nucleophiles add reversibly to the C=O group, giving an opportunity for any direct addition product to revert to starting materials and react again.



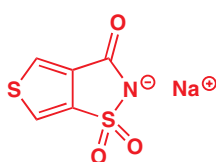
Regioselectivity in action

We finish with an example that illustrates several aspects of chemoselectivity as well as introducing the subjects of the next two chapters. The first synthetic sweetener was saccharin but newer ones such as the BASF compound thiophenesaccharin are much in demand. The sodium salt is the active sweetener but the neutral compound has to be made via the simpler intermediate thiophene.

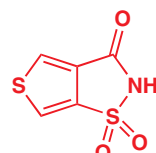
■ Thiophene is the name for this sulfur-containing aromatic compound. There is more about it in Chapter 29.



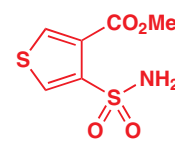
saccharin



salt of thiophenesaccharin



thiophenesaccharin



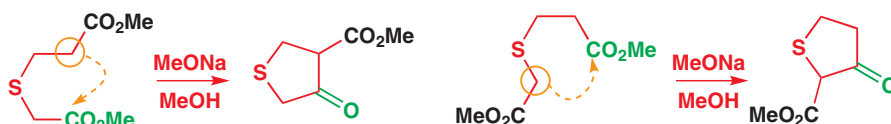
intermediate thiophene

The synthesis started with a conjugate addition of a thiol to an unsaturated ester. The thiol is obviously the nucleophile and regioselectively chooses conjugate addition rather than attack on either ester group.

■ The thiol *could* attack its own ester group, leading to polymerization, but it doesn't.



In the next step the diester is treated with base and a carbonyl condensation reaction occurs of the type you will meet in Chapter 26. There is a real question of regioselectivity here: an enolate could form next to either ester (as shown by the orange circles) and would then attack the other ester as a nucleophile. There is little to choose between these alternatives but the first was wanted and was selected by careful experimentation, although only in 50% yield. This was acceptable on a large scale as the product could be separated by crystallization, the most practical of all methods.



Reactions such as this—the attack of enolates on carbon electrophiles—form the subject of the next two chapters, where we will discuss in detail the mechanism of this type of reaction.

Further reading

There is a basic introduction in S. Warren and P. Wyatt, *Organic Synthesis: the Disconnection Approach*, Wiley, Chichester, 2008, chapter 3.

Ortholithiation: P. Wyatt and S. Warren, *Organic Synthesis: Strategy and Control*, Wiley, Chichester, 2007. J. Clayden, *Organolithiums: Selectivity for Synthesis*, Pergamon, 2002.

Reduction of nitro groups: L. McMaster and A. C. Magill, *J. Am. Chem. Soc.*, 1928, **50**, 3038. Bromination of nitrobenzene: B. S. Furniss, A. J. Hannaford, P. W. G. Smith, and A. T. Tatchell, *Vogel's Textbook of Practical Organic Chemistry*, Longman, 5th edn, 1989, p. 864.

Formation of diazonium salts and conversion into aryl halides: B. S. Furniss, A. J. Hannaford, P. W. G. Smith, and A. T. Tatchell, *Vogel's Textbook of Practical Organic Chemistry*, Longman, 5th edn, 1989, pp. 933, 935. Iodolactonisation: B. S. Furniss, A. J. Hannaford, P. W. G. Smith, and A. T. Tatchell, *Vogel's Textbook of Practical Organic Chemistry*, Longman, 5th edn, 1989, p. 734.

The Wittig-style Horner-Wadsworth-Emmons alkene synthesis: W. S. Wadsworth and W. D. Emmons, *Org. Synth. Coll.*, 1973, **5**,

547. P. Wyatt and S. Warren, *Organic Synthesis: Strategy and Control*, Wiley, Chichester, 2007. Synthesis of non-conjugated compounds: C. W. Whitehead, J. J. Traverso, F. J. Marshall, and D. E. Morrison, *J. Org. Chem.*, 1961, **26**, 2809.

Regioselective electrophilic attack on dienes: R. B. Moffett, *Org. Synth. Coll.*, 1963, **4**, 238. K. Nakayama, S. Yamada, H. Takayama, Y. Nawata, and Y. Itaka, *J. Org. Chem.*, 1984, **49**, 1537.

Buffered epoxidation to avoid rearrangement of product: M. Imuta and H. Ziffer, *J. Org. Chem.*, 1979, **44**, 1351. Mono- and di-epoxidation of dienes: M. A. Hashem, E. Manteuffel, and P. Weyerstahl, *Chem. Ber.*, 1985, **118**, 1267.

Regioselective bromination of dienes: A. T. Blomquist and W. G. Mayes, *J. Org. Chem.*, 1945, **10**, 134. Regioselective nucleophilic substitution on allylic bromides: A. C. Cope, L. L. Estes, J. R. Emery, and A. C. Haven, *J. Am. Chem. Soc.*, 1951, **73**, 1199. V. H. Heasley and P. H. Chamberlain, *J. Org. Chem.*, 1970, **35**, 539. But ignore the theoretical part especially the three 'different' intermediates.

Check your understanding



To check that you have mastered the concepts presented in this chapter, attempt the problems that are available in the book's Online Resource Centre at <http://www.oxfordtextbooks.co.uk/orc/clayden2e/>